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Smart Crop Irrigation Prediction Using Machine Learning

SIC – AI802

Agenda

- Team Members
- Motivation & Problem
- Machine Learning Solution
- System Workflow
- Data Overview
- Data Exploring and Preprocessing
- Exploratory Data Anlaysis
- Models Training
- Models Evaluation
- Business Insights and Recommendation
- Limitation and Future Work

Team Members

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Motivation & Problem

Motivation & Problem

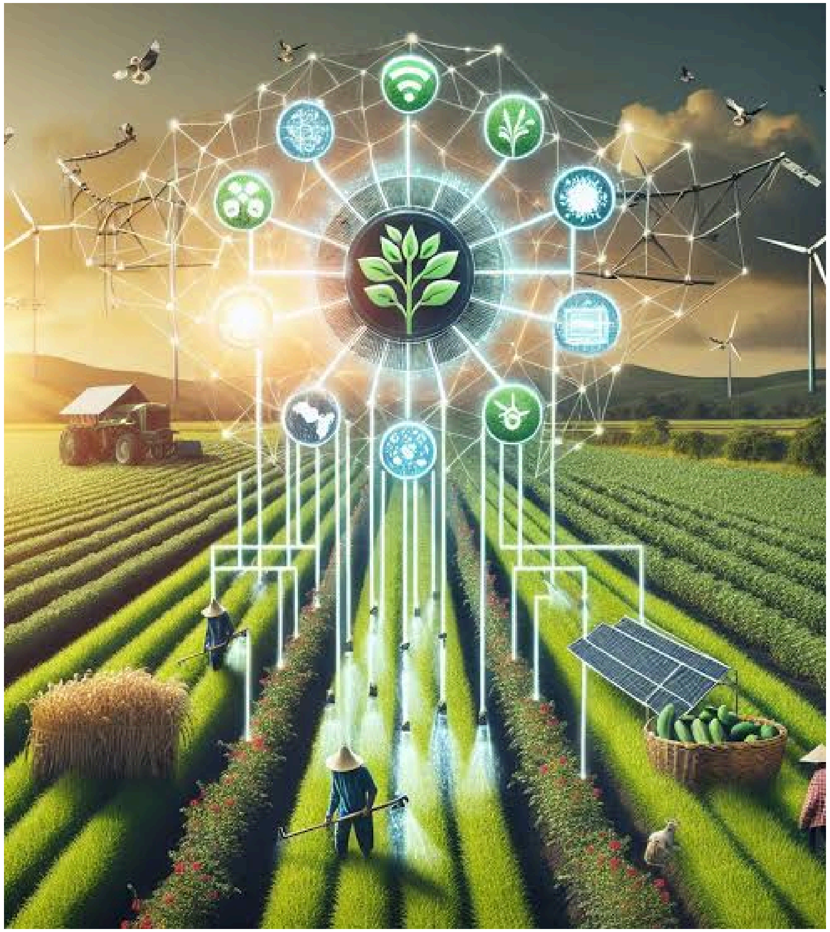
- Agriculture depends heavily on effective water management, yet irrigation decisions are often challenging because crop water requirements vary with soil and environmental conditions. Improper irrigation can result in water stress and negatively affect crop growth, while unnecessary irrigation contributes to inefficient water use.

- The challenge is to identify, from measurable environmental and crop conditions, whether a crop currently requires irrigation.

Machine Learning Solution

Crop Irrigation Detection

- We developed a machine learning–based classification system that analyzes crop and environmental conditions to predict whether irrigation is required.
- The model learns from historical crop data and identifies patterns between soil moisture, temperature, humidity, soil type, and growth stage to support data-driven irrigation decisions.



System Workflow

Our System Workflow

1_Dataset Overview

2-Dataset Exploring and Preprocessing

3_Exploratory Data Analysis

4_Models Training

5_Model Evaluation

6_Business Insights and Recommendations

Dataset Overview

Dataset Overview

Smart Agriculture Dataset

16,411 records of crop-related data.
Contains environmental factors that affect crop growth and irrigation needs.

Features:
Crop ID: Unique identifier for each crop sample.
Soil Type: Type of soil, such as Black Soil or Red Soil.
Seedling Stage: Current growth stage of the crop.
MOI (Moisture Index): Indicates the soil moisture level.
Temperature: Ambient temperature in °C.
Humidity: Relative humidity percentage.
Result: Irrigation requirement → 1 = Yes, 0 = No

Data Exploring and Preprocessing

Data Exploring and Preprocessing I

- **Data Cleaning & Filtering:**

- **Handling Duplicates**: Removed duplicate entries to prevent data leakage (16,411 to 16,283 records).
- **Invalid Class Removal**: Filtered out erroneous target entries (result != 2), leaving 15,161 clean samples.
- **Outlier Detection:** Checked numerical features (moi, temp, humidity) using IQR bounds, confirming 0% extreme outliers.

- **Feature Engineering & Transformation:**

- **Growth Stage Mapping:** Converted Seedling Stage text into ordinal numeric values (growth_stage_order from 1 to 8).
- **Interaction Features:** Created combined ratios moi_temp_ratio (moi / temp) and moi_humidity_index (moi * humidity / 100).

Data Exploring and Preprocessing II

- **Column Normalization:** Standardized column naming conventions for consistent processing.

- **Pipeline Preparation & Splitting:**

- **Stratified Data Split**: Split dataset into 80% Training and 20% Testing using stratify=y to preserve target balance.

- **ColumnTransformer Pipeline:** Applied StandardScaler to numerical features and OneHotEncoder to categorical features (crop_name, soil_type).

Exploratory Data Analysis

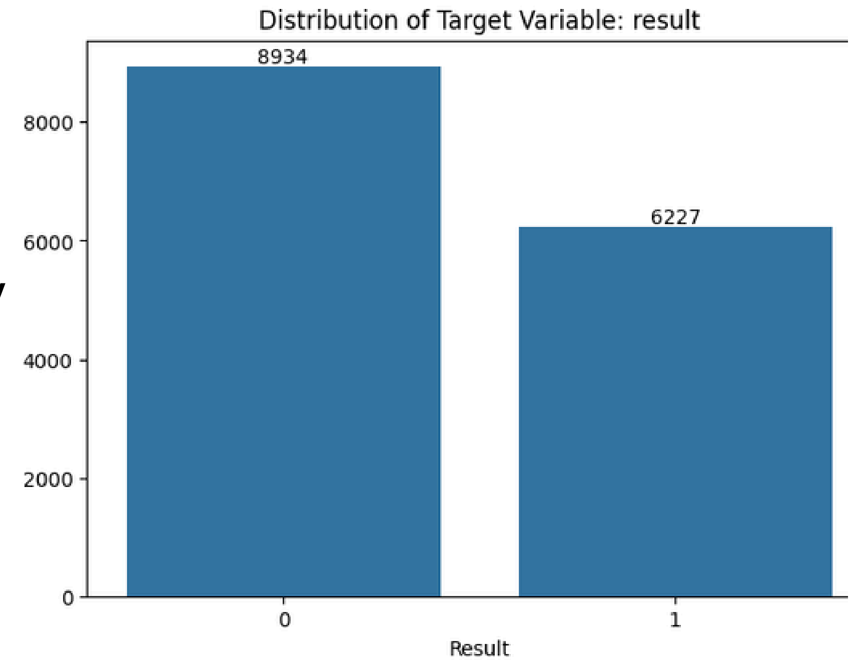
EDA

Exploratory Data Analysis (EDA)

- 15,161 observations & 10 features
- Numerical + categorical variables
- No missing values
- Target: result (binary)
- Categorical: Crop, Soil Type, Growth Stage
- Numerical: Moisture, Temperature, Humidity engineered features

- **Target Distribution**

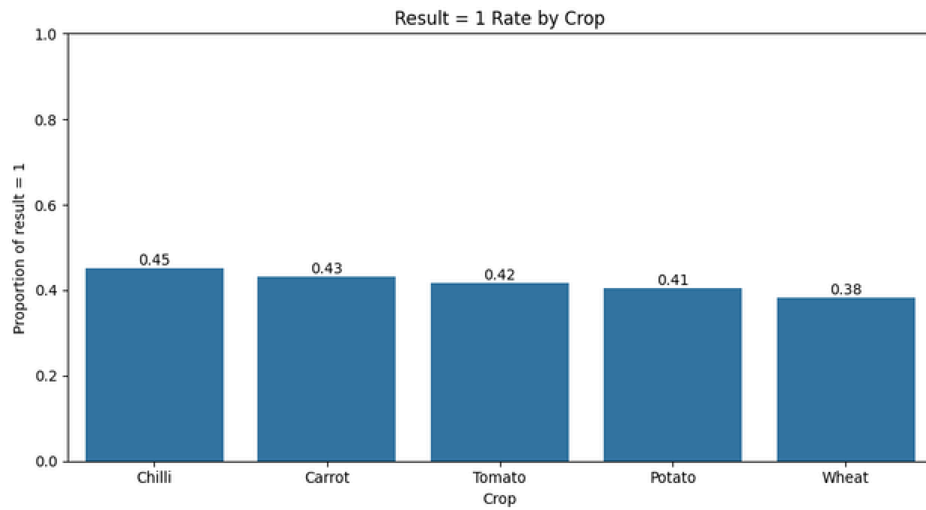
- Result = 0 → 58.93%
- Result = 1 → 41.07%

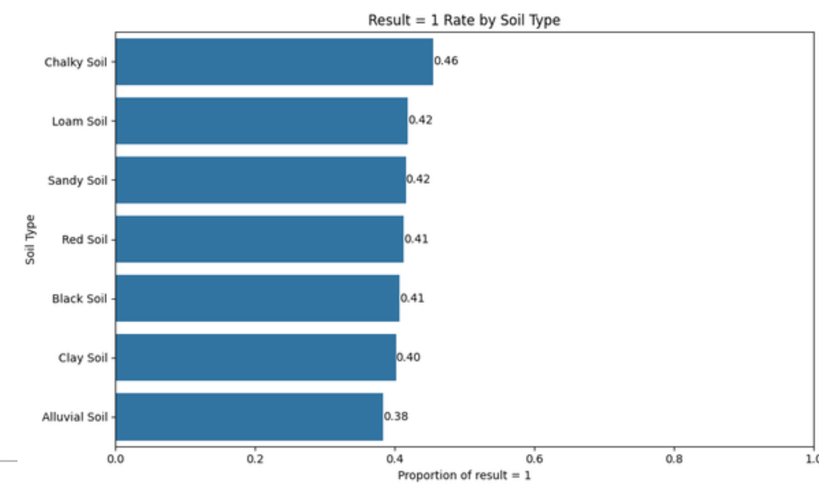


EDA

Key EDA Findings

- **1. Crop-wise differences**
- Chilli has the highest Result = 1 rate (45.22%).
- Wheat has the lowest rate (38.31%).
- **2. Soil-wise differences**
- Chalky Soil shows the highest Result = 1 rate (45.57%).
- Alluvial Soil shows the lowest rate (38.32%).
- **3. Growth stages**
- Result rates vary across growth stages.
- The highest rates occur during Fruit/Grain/Bulb Formation and Maturation (43.67%).
- The lowest rate occurs during Pollination (39.19%).
- Crop, soil type, and growth stage show differences in Result = 1 rates and may provide useful predictive information.





EDA

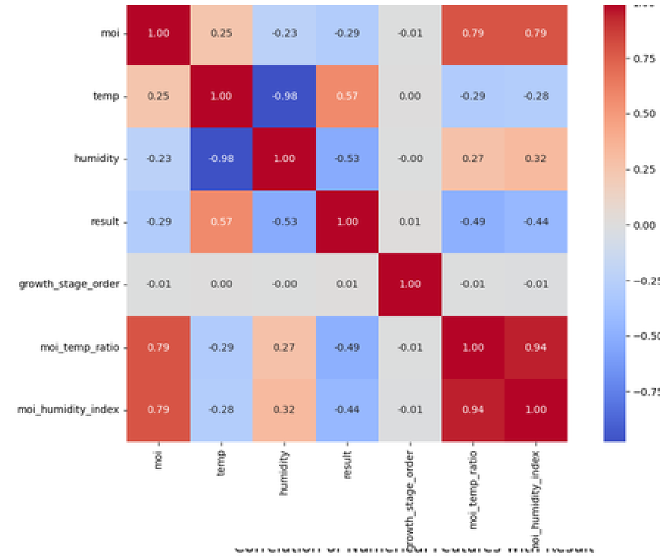
Numerical Features & Correlations
Correlation with Result:

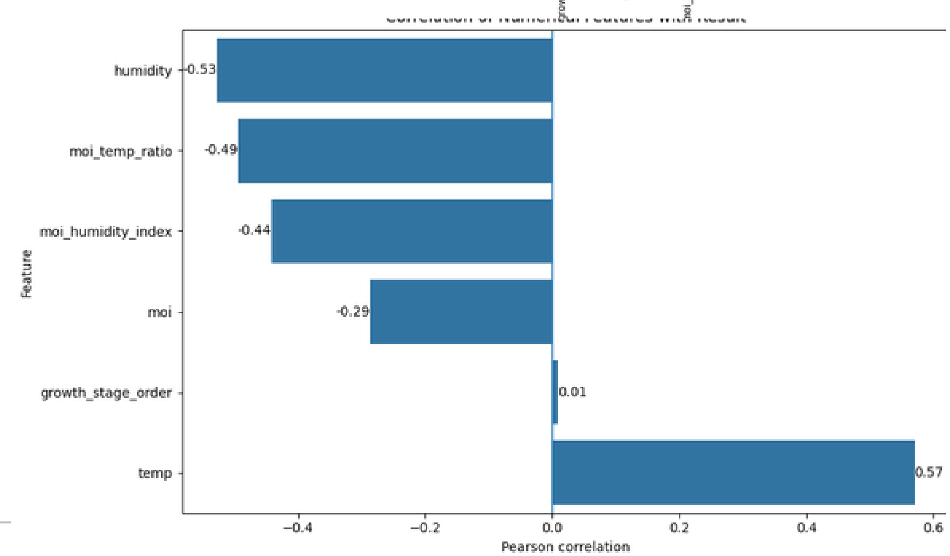
- Temperature: +0.57
- Humidity: −0.53
- Moisture/Temperature Ratio: −0.49
- Moisture/Humidity Index: −0.44
- Moisture: −0.29
- Growth Stage Order: +0.01

Key Insight

Temperature and humidity show the strongest linear associations with the target.

Correlation indicates association, not causation.





Model Training

Model Training

Model Training Setup

- Logistic Regression: Linear Baseline
- KNN: Distance-Based Classifier (k=5)
- Decision Tree: Interpretable Tree Baseline
- Random Forest: Bagging Ensemble (100 Trees)
- XGBoost: Gradient Boosted Trees (100 Trees)
- SVM (RBF): Kernel-Based Classifier
-

Key Training Details:

- Trained on preprocessed & feature-engineered data (X_train_transformed).
- Evaluated on unseen test set (X_test_transformed).
- Fixed seed (random_state=42) used across all models for reproducibility.

Model Evaluation

Model Evaluation I

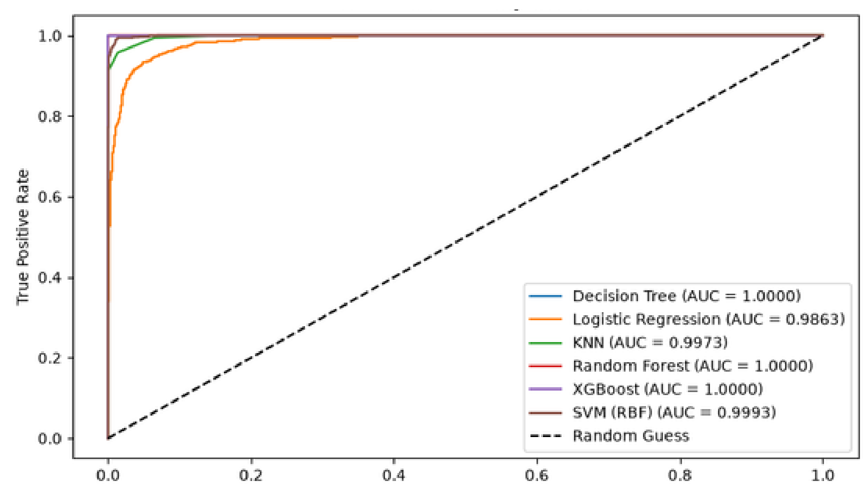
1. Test Performance Comparison

- **Top Performers:** Decision Tree, XGBoost, and Random Forest achieve near-perfect metrics (Accuracy: 99.80% – 100.00%).
- **Zero Crop Hazard:** Decision Tree, Random Forest, and XGBoost reach a 100% Recall (1.0000), ensuring no necessary irrigation events are missed.
- **Baseline Lag:** Logistic Regression (94.26%) and KNN (97.46%) lag behind, confirming the necessity of non-linear decision boundaries.

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.9426	0.9489	0.9093	0.9287	0.9863
1	K-Nearest Neighbors (KNN)	0.9746	0.9803	0.9575	0.9687	0.9973
2	Decision Tree	1.0000	1.0000	1.0000	1.0000	1.0000
3	Random Forest	0.9980	0.9952	1.0000	0.9976	1.0000
4	XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000
5	SVM (RBF)	0.9842	0.9942	0.9671	0.9805	0.9993

2. Overfitting & Generalization

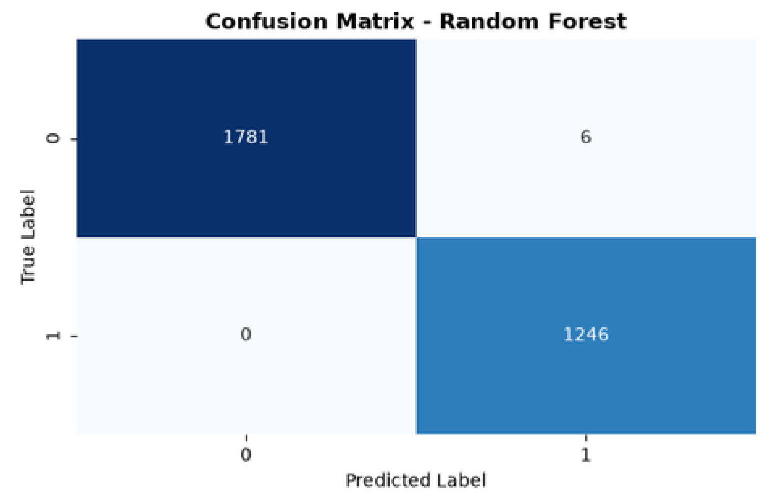
- **High Stability:** All models exhibit minimal generalization gaps (< 1.67%).
- **Model Reliability:** Decision Tree and XGBoost achieve identical train/test scores **(1.0000),**
- while Random Forest maintains an ideal generalization gap (0.20%), avoiding memorization risk.



Model Evaluation II

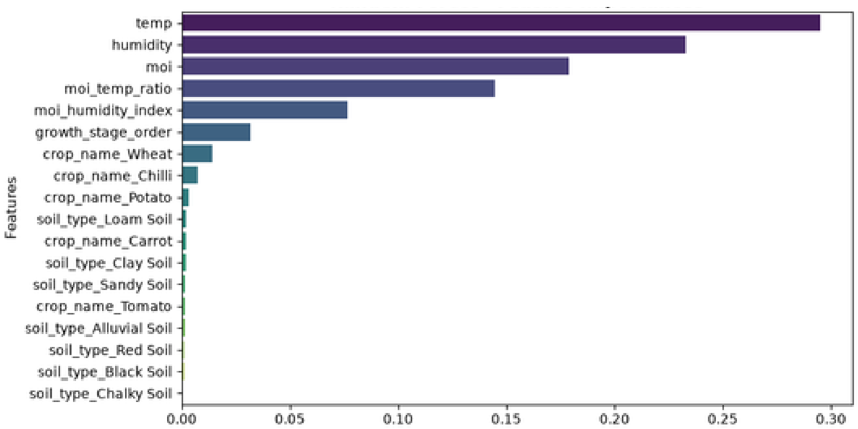
3. Confusion Matrix Analysis (Random Forest)

- **Zero False Negatives (FN = 0):** Eliminates crop wilting risk by capturing every single irrigation requirement.
- **Minimal False Positives (FP = 6):** Out of 3,027 test samples, only 6 unnecessary irrigation events occurred, minimizing resource waste.



. Feature Importance Insights

- **Primary Environmental Drivers:** temp, humidity, and moi account for over 60% of total predictive power.
- **Engineered Ratios:** Features like moi_temp_ratio and moi_humidity_index rank highly, validating feature engineering impact.
- **Low Categorical Weight:** Static crop and soil categories contribute minimally compared to real-time environmental conditions.



Business Insights and Recommendation

Business Insights and Recommendation

Business Recommendations

- **Deploy Random Forest to IoT Edge Nodes:** Integrate the trained Random Forest model into smart agricultural controllers to automate real-time valve control based on dynamic sensor feeds.
- **Prioritize Real-Time Weather Sensors:** Investment should focus on high-precision temperature, humidity, and soil moisture sensors, as these three variables account for over 60\% of predictive decision weight.
- **Dynamic Safety Thresholds:** Utilize predicted probability thresholds rather than strict binary outputs to trigger early warning alerts before severe soil dehydration occurs.
- **Optimize Resource Efficiency:** Automating irrigation based on model predictions prevents unnecessary watering during high-humidity or low-temperature periods, directly reducing water waste and energy expenditure.

Future Enhancements

Future Enhancements

Limitations & Future Work

- **Data Constraints:** The current dataset relies on static categorical soil/crop labels and simulated environmental conditions without temporal/time-series continuity.
- **Risk of Overfitting / Memorization:** Perfectly separated decision boundaries in **Decision Tree and XGBoost** (100\% accuracy) indicate potential high variance or data redundancy that requires validation on broader, real-world field environments.

Future Work & Expansion:

  - **Time-Series Integration:** Collect continuous sequential sensor data to train Recurrent Neural Networks (LSTM/GRU) or Temporal Transformers for predictive forecasting.
  - **Real-World Deployment & A/B Testing:** Field-test the pipeline using physical microcontroller units (e.g., ESP32/Raspberry Pi) and measure real water savings against manual farming schedules.
  - **Hyperparameter Tuning & Compression:** Apply model quantization and pruning to lightweight ensemble models to enable microsecond-level inference on low-power IoT microcontrollers.

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